

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

Course No: MTE 3101

Course Title: Control Systems Sessional

Experiment No: 03

Experiment Name: Design of a PID Controller for the systems by using trial and error method and analysis their performance

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Date of Submission:

26th April, 2025

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Experiment Name: Design of a PID Controller for the systems by using trial and error method and analysis their performance.

Objectives:

1. To design PID, PD & PI controllers for given systems
2. To analyze and compare the performance of PID, PD & PI controllers through simulation
3. To design a PID controller for the given systems using the trial-and-error method and to analyze and compare its performance characteristics

Theory:

PID control, representing proportional-integral-derivative control, is a feedback mechanism in control system, often referred to as three-term control. By adjusting three parameters—the proportional, integral, and derivative values of a process variable's deviation from its set point—specific control actions are effectively tailored.[1]

Proportional (P) response: The proportional gain (K_c) determines the ratio of output response to the error signal. In general, increasing the proportional gain will increase the speed of the control system response.

Integral (I) response: The integral component sums the error term over time. The result is that even a small error term will cause the integral component to increase slowly. The integral response will continually increase over time unless the error is zero, so the effect is to drive the Steady-State error to zero.

Derivative(D) response: The derivative component causes the output to decrease if the process variable is increasing rapidly. The derivative response is proportional to the rate of change of the process variable.[2]

Some control actions only require two of the PID controller's parameters, setting the third to zero. This flexibility allows PID controllers to operate as PI (proportional-integral), PD (proportional-derivative), or simply P or I. The proportional, derivative and integral parameters can be expressed as $-K_p$, K_d and K_i . All these three parameters have an effect on the closed loop control system. It affects rise time, settling time and overshoot and also the Steady state error.[1]

PID Controller Equation:

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

In Laplace Domain,

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

Control Response	Rise time	Settling time	Overshoot	Steady state error
K_p	decrease	small change	increase	decrease
K_d	small change	decrease	decrease	no change
K_i	decrease	increase	increase	eliminate

Table 1: PID Parameters Summary Table [1]

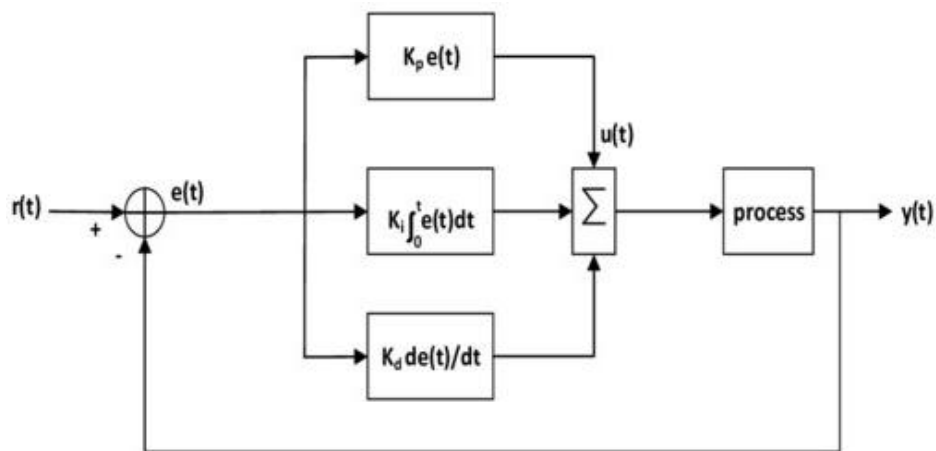


Figure 1: Basic PID controller block diagram. [3]

Required Apparatus:

MATLAB

System Models:

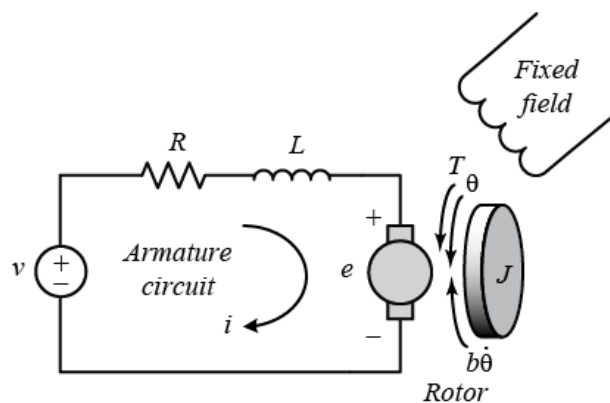


Figure 2: System Representation of DC Motor

Transfer Function:

Motor system transfer function,

$$\frac{\theta(s)}{v(s)} = \frac{K}{\{(Js + b)(Ls + R) + K^2\}}$$

MATLAB Code:

```
clc;
clear all;
clf;

%system Parameters
J=0.01;
b=0.1;
K=0.01;
R=1;
L=0.5;
s= tf('s');

%Transfer function Representation
TF=K/((J*s+b)*(L*s+R)+K^2);

%Design of PID Controller
Kp=274.3563;
Ki=816.9301;
Kd=23.0348;
PID=pid(Kp,Ki,Kd);

%Design of PI Controller
Kp1=22.9417;
Ki1=42.0116;
Kd1=0;
PI=pid(Kp1,Ki1,Kd1);

%Design of PD Controller
Kp2=386.3211;
Ki2=0;
Kd2=21.849;
PD=pid(Kp2,Ki2,Kd2);

%Closed Loop system
CL1=feedback(TF*PID,1);
CL2=feedback(TF*PI,1);
CL3=feedback(TF*PD,1);

%Outputs

figure(1)
step(TF);
hold on
step(CL1);
hold on
step(CL2);
hold on
step(CL3);
hold on

sys=ss(TF)

stepinfo(CL1)
stepinfo(CL2)
stepinfo(CL3)
```

State-Space Model:

$$A = \begin{matrix} & x1 & x2 \\ x1 & -12 & -5.005 \\ x2 & 4 & 0 \end{matrix}$$

$$B = \begin{matrix} & u1 \\ x1 & 0.5 \\ x2 & 0 \end{matrix}$$

$$C = \begin{matrix} & x1 & x2 \\ y1 & 0 & 1 \end{matrix}$$

$$D = \begin{matrix} & u1 \\ y1 & 0 \end{matrix}$$

Simulink:

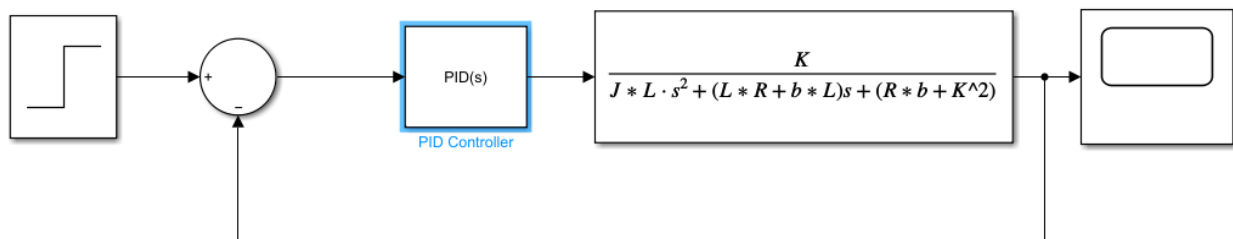


Figure 3: PID Control System Block

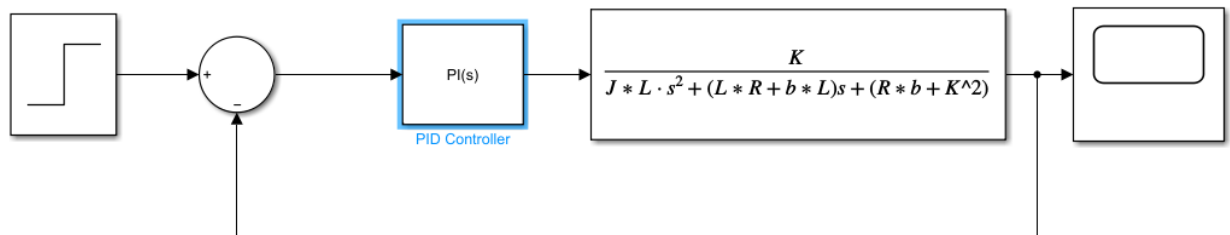


Figure 4: PI Control System Block

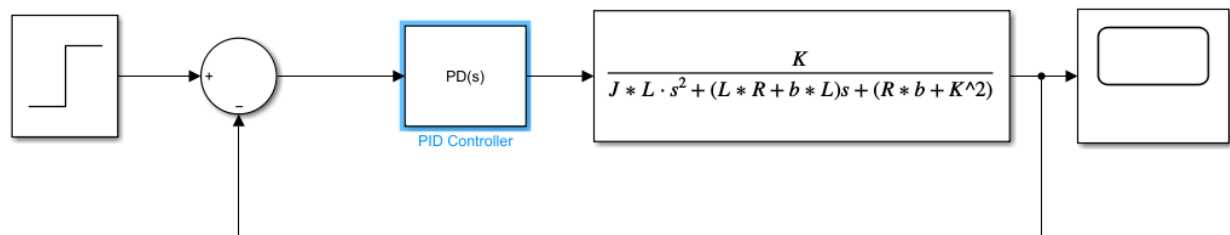


Figure 5: PD Control System Block

Result:

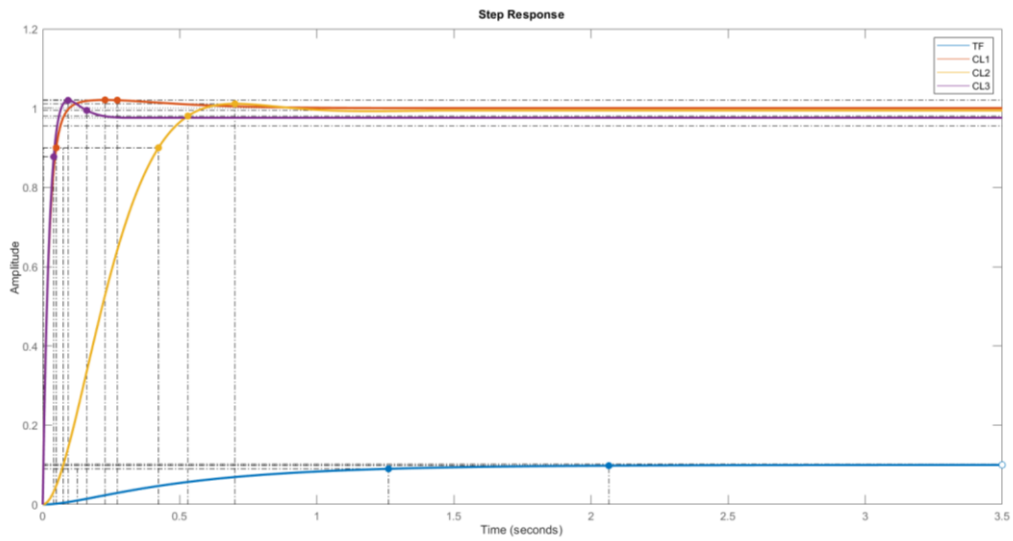


Figure 6: System Response with PID, PD & PI Controller

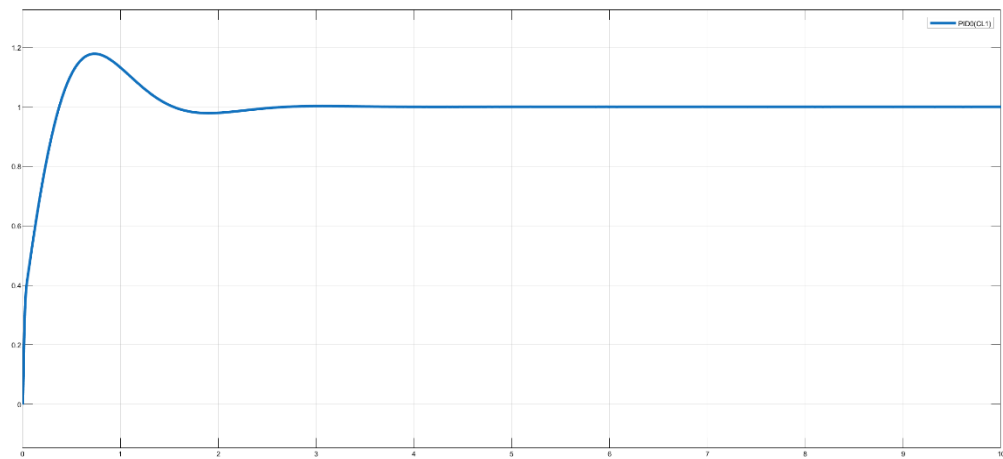


Figure 7: System Response with PID Controller (Simulink)

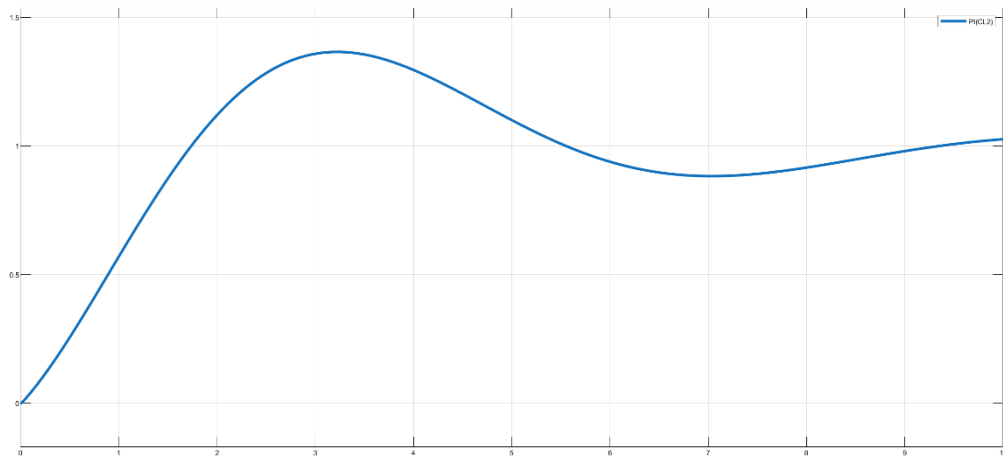


Figure 8: System Response with PI Controller (Simulink)

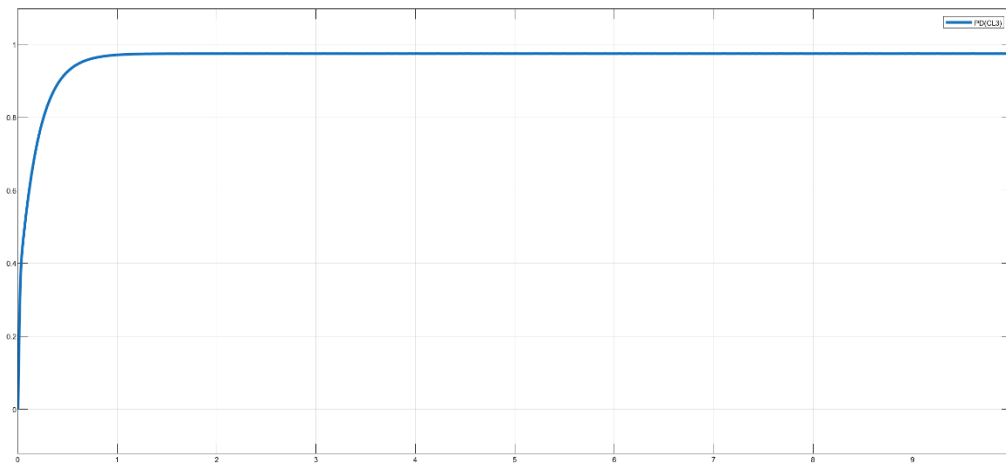


Figure 9: System Response with PD Controller (Simulink)

Performance Analysis:

Controller	Rise Time	Settling Time	Peak Time	Overshoot (%)	Peak Value
PID	0.0470	0.2721	0.2272	2.0633	1.0206
PI	0.3472	0.5293	0.7011	1.0958	1.0110
PD	0.0379	0.1604	0.0926	4.6029	1.0196

Discussion & Conclusion:

The performance analysis shows that each controller has specific advantages. The PID controller has a balanced performance with a moderate rise time, settling time, and low overshoot, making it excellent for systems that require fast but reliable responses. While having zero overshoot, the PI controller has the slowest rise and settling durations, indicating that it is best suited to applications where overshoot is carefully avoided. The PD controller has the fastest rise and settling periods, with little overshoot, but slightly lower peak value stability. Overall, the PID controller offers the optimum balance of speed and accuracy, whereas PI and PD controllers are better suited for applications that prioritize overshoot removal or response speed, respectively.

Reference:

- [1] Electrical4U, "PID control – proportional integral derivative," *Electrical4U*, [Online]. Available: <https://www.electrical4u.com/pid-control/>. Accessed: Apr. 26, 2025
- [2] National Instruments, "PID Theory Explained," *NI*, [Online]. Available: <https://www.ni.com/en/shop/labview/pid-theory-explained.html>. Accessed: Apr. 26, 2025.
- [3] M. Malekabadi, M. Haghparast, and F. Nasiri, "Air Condition's PID Controller Fine-Tuning Using Artificial Neural Networks and Genetic Algorithms," *Computers*, vol. 7, no. 2, p. 32, May 2018, doi: 10.3390/computers7020032.